# Rna motivation

RNA is made up of strands of four different bases that encode genomic information in specific ways. The bases are elements of the set $B = \{\mathtt{A}, \mathtt{C}, \mathtt{U}, \mathtt{G}\}$. Strands are ordered nonempty finite sequences of bases.

Formally, to define the set of all RNA strands, we need more than roster method or set builder descriptions.

# Recursive sets definition

**New! Recursive Definitions of Sets**: The set $S$ (pick a name) is defined by:

| | |
|---|---|
| Basis Step: | Specify finitely many elements of $S$ |
| Recursive Step: | Give rule(s) for creating a new element of $S$ from known values existing in $S$, and potentially other values. |

The set $S$ then consists of all and only elements that are put in $S$ by finitely many (a nonnegative integer number) of applications of the recursive step after the basis step.

CC BY-NC-SA 2.0 Version October 12, 2026 (1)

# Set recursive examples

**Definition** The set of nonnegative integers $\mathbb{N}$ is defined (recursively) by:

Basis Step:
Recursive Step:

Examples:

**Definition** The set of all integers $\mathbb{Z}$ is defined (recursively) by:

Basis Step:
Recursive Step:

Examples:

**Definition** The set of RNA strands $S$ is defined (recursively) by:

| | |
|---|---|
| Basis Step: | $\mathtt{A} \in S, \mathtt{C} \in S, \mathtt{U} \in S, \mathtt{G} \in S$ |
| Recursive Step: | If $s \in S$ and $b \in B$, then $sb \in S$ |

where $sb$ is string concatenation.

Examples:

**Definition** The set of bitstrings (strings of 0s and 1s) is defined (recursively) by:

Basis Step:
Recursive Step:

*Notation:* We call the set of bitstrings $\{0,1\}^*$ and we say this is the set of all strings over $\{0,1\}$.

Examples:

CC BY-NC-SA 2.0 Version October 12, 2026 (2)